

Yumeng He

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EDUCATION

University of Southern California

Master of Science in Computer Science

Aug 2024 – May 2026 (Expected)

Los Angeles, CA, USA

- Cumulative GPA: 3.83/4.0
- Relevant Course: Computer Graphics, Computer Animation and Simulation, 3D Graphics and Rendering

University of Toronto

Honours Bachelor of Science degree with High Distinction

Sep 2019 – May 2024

Toronto, ON, Canada

- Computer Science and Mathematics double major
- Dean's List Scholar for 4 semesters
- Cumulative GPA: 3.71/4.0, highest Sessional Average: 92.40%.
- Relevant Courses: Linear Algebra I, Linear Algebra II, Intro Machine Learning, Intro Artif Intell

WORKING EXPERIENCE

Software Developer Intern | full-time

August 2022 – August 2023

HCL Canada Inc.

Toronto, ON, Canada

- Utilized Jenkins to automate build and deployment processes and updated existing company code to address technical debt, significantly accelerating production releases and reducing human error.
- Built Docker images on both xLinux and pLinux platforms, ensuring consistent and scalable deployments.
- Integrated SonarQube into the development pipeline, documented a step-by-step usage guide, presented findings to cross-functional teams, and resolved 3 critical bugs plus 308 code smells (4 blockers, 41 major, 266 info).
- Developed a shell script to automatically clean up old logs every Sunday at 00:00 EST, preventing clutter and improving system performance.

RELEVANT PROJECTS

Incompressible Fluid Simulation: A Comparison | C++, OpenGL

Spring 2025

- A 2D incompressible fluid simulation implemented in C++ with visualization using OpenGL and GLUT.
- The main objective is to compare different fluid simulation methods, including: Grid-based (Stable Fluids), Particle-based (SPH), Particle-In-Cell (PIC), hybrid PIC/FLIP method(PIC/FLIP), Affine Particle-In-Cell (APIC).

Collision detection with penalty method and IPC | C++

Spring 2025

- Simulated a jello cube using a 3D mass-spring system.
- Modeled a network of Structural, Shear, and Bend springs to capture deformation behavior.
- Applied the penalty method for collision detection against boundaries and obstacles.
- Additionally implemented a simplified Incremental Potential Contact (IPC) method with line search and continuous collision detection.

Ray Tracing | C++

Fall 2024

- Developed a robust ray tracer in C++ capable of rendering both triangles and spheres using Phong shading, shadow rays, and recursive reflections for mirror-like surfaces.
- Implemented supersampling antialiasing, soft shadows from area lights, and the Möller–Trumbore algorithm for efficient ray-triangle intersections, enhancing visual realism and performance.

Simulating a Roller Coaster | OpenGL

Fall 2024

- Implemented a first-person roller coaster simulator in C++ and OpenGL, harnessing Catmull-Rom splines for realistic track geometry and smooth camera orientation.
- Created multi-shader pipelines (texture mapping for ground, Phong shading for rails) with per-pixel lighting to achieve immersive, real-time visual effects.
- Added user-controlled speed, automated screenshot capture, and custom rail cross-sections for a fully interactive 3D experience.

TECHNICAL SKILLS

Languages/Tools: Proficient in C++ and Python (with Jupyter), plus JavaScript (HTML/CSS), MySQL, Docker

Graphics Experience: Developed real-time rendering pipelines (OpenGL, shaders) for interactive 3D projects

Additional: Node.js, React, Gradle, Ant, R, Shell scripting

Communication: Fluent in English and Mandarin